

Formative R activities: Health Data Analysis and Hypothesis Testing

1. Objective

This explanatory analysis covers statistical activities conducted in R using the provided **Health Data** set. The dataset contains 210 observations and 17 variables, including age, sex, religion, occupation, income, and medical conditions such as blood pressure, peptic ulcer, and diabetes; pre- and post-test data; and admission and discharge data.

The activities focused on data exploration, data visualisation and hypothesis testing in R. The R reporting script is available in the Appendix section.

2. Dataset Exploration

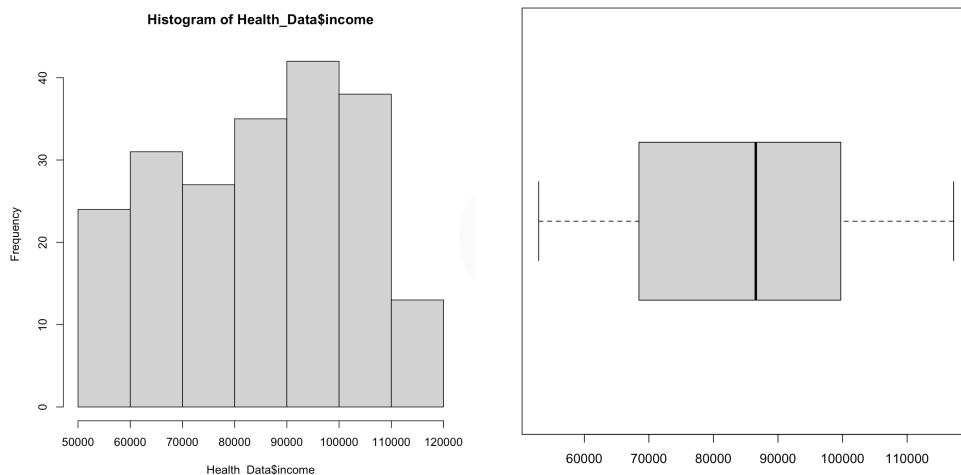
2.1. Mean, Median, Mode

Variable	Mean	Median	Mode
sbp	127.7333	123	[1] 120 attr(,"freq") [1] 12
dbp	82.76667	82	[1] 74 80 82 attr(,"freq") [1] 13
income	85194.49	86560.5	No mode

2.2. Five-figure summary - Monthly income (income)

Min.	1st Qu.	Median	3rd Qu.	Max.
52933	68636	86560	99696	117210

2.3. Visualisation - Monthly income (income)



The histogram shows a slightly left-skewed distribution. The box plot shows that the median is pretty centred, indicating no strong skewness. The left and right whiskers are of similar length, indicating that the distribution is approximately symmetric. There are also no obvious outliers visible in this group.

3. Hypothesis Testing

The objective of the hypothesis test was to determine whether there is an association between systolic blood pressure and the presence or absence of a peptic ulcer.

3.1. Variables Characteristic

Given that these independent variables are numerical and categorical, they enable Hypothesis testing via the **Independent Two-Sample t-Test**, allowing us to evaluate whether the mean systolic blood pressure (sbp) is the same for patients with and without a peptic ulcer condition.

- systolic blood pressure - *numeric variable* (`num [1:210]`)
- peptic ulcer - *categorical variable* (`chr [1:2] "Yes" "No"`)

3.2. Hypothesis

H₀: Mean systolic blood pressure (sbp) is the same for patients with and without peptic ulcer.

H₀: $\mu_1 = \mu_2$

H1: Mean systolic blood pressure (sbp) differs between patients with and without peptic ulcer.

H1: $\mu_1 \neq \mu_2$

3.3. Normality Testing - Shapiro.test

The Shapiro-Wilk normality test yielded p-value = 3.345e-06, which is > 0.05 ; hence, we fail to reject H_0 , indicating that the income follows a normal distribution.

3.4. Independent Two-Sample t-Test

The Welch Two-Sample t-Test results did not reveal a statistically significant difference in mean systolic blood pressure (SBP) between patients with and without a peptic ulcer.

Test Results	
t = 1.2142	t is the t-test statistic value
df = 57.562	df is the degrees of freedom
p-value = 0.2296	p-value is the significance level of the t-test (p-value = 0.00)
conf.int = [-2.889367 11.795703])	conf.int is the confidence interval of the mean at 95%
mean = 131.3171, 126.8639	Sample estimates are the mean values of the sample.
	The p-value of the test is 0.2296, which is higher than the significance level $\alpha = 0.05$. We fail to reject H_0. We can conclude that the Mean systolic blood pressure (SBP) is the same in patients with and without a peptic ulcer. H0: $\mu_1 = \mu_2$

Appendix:

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> # Unit 7: Health_Data_Activities and Hypothesis Testing
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> ##
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```
> #View(Health_Data)
```

```
> #summary(Health_Data)
```

```
> #sapply(Health_Data, class)
```

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> ##
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```
> # Mean, Median, Mode for sbp, dbp and income
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```
> ##
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```
> library(DescTools)
```

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```

```
> #Data characteristic of Systolic BP (sbp)
```

```
> mean(Health_Data$sbp)
```

```
[1] 127.7333
```

```
> median(Health_Data$sbp)
```

```
[1] 123
```

```
> Mode(Health_Data$sbp)
```

```
[1] 120
```

```
attr("freq")
```

```
[1] 12
```

```
>
```

```
> #Data characteristic - Diastolic BP (dbp)
```

```
> mean(Health_Data$dbp)
```

```
[1] 82.76667
```

```
> median(Health_Data$dbp)
```

```
[1] 82
```

```
> Mode(Health_Data$dbp)
```

```
[1] 74 80 82
```

```
attr("freq")
```

```
[1] 13
```

```
>
```

```
> #Data characteristic - Monthly income (income)
```

```
> mean(Health_Data$income)
```

```
> # Normality Testing - shapiro.test
```

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> ##
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```

```
> shapiro.test(Health_Data$sbp)
```

Shapiro-Wilk normality test

data: Health_Data\$sbp

W = 0.95474, p-value = 3.345e-06

```
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```
> ##
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```
> # Hypothesis Testing: Independent Two-Sample t-Test
```

```
> # H0: Mean systolic blood pressure (sbp) is the same for patients with and without peptic ulcer condition.
```

```
> # H1: Mean systolic blood pressure (sbp) differs between patients with and without peptic ulcer condition
```

```
> ##
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```
> t.test(sbp~pepticulcer,data = Health_Data)
```

Welch Two Sample t-test

data: sbp by pepticulcer

t = 1.2142, df = 57.562, p-value = 0.2296

alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0

95 percent confidence interval:

-2.889367 11.795703

sample estimates:

mean in group 1 mean in group 2

131.3171 126.8639